

Sunil Tiwari

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EDUCATION

University of Utah

Master of Science in Cybersecurity Management (Specialization: Agentic AI in SecOps)

Salt Lake City, UT

Expected July 2026

Utah Valley University

Bachelor of Science in Information Technology (Network Security Focus)

Orem, UT

December 2022

TECHNICAL SKILLS

Agentic AI & Orchestration: CrewAI | LangChain | LLM SecOps | Prompt Engineering | Llama 3.2 | Ollama | RAG

Security Engineering: Autonomous Incident Response | SIEM (Splunk/Wazuh) | Forensic Triage | MITRE ATT&CK | Linux Kernel Networking | Vulnerability Management

Infrastructure & DevOps: NVIDIA DGX (ARM64) | Python (FastAPI) | Docker | Azure | Active Directory | PowerShell | Terraform

ENGINEERING PROJECTS

Hybrid-LLM Autonomous SOC & Active Defense Framework

December 2025 – Present

Security Automation Architect

- Architected a Hybrid-LLM Autonomous SOC utilizing the CrewAI framework to orchestrate local edge models (Llama 3.2) and cloud intelligence (GPT-4o) for zero-intervention threat triage and incident response.
- Integrated isolated Docker honeypots with a custom FastAPI backend, engineering regex-driven Python telemetry bridges to dynamically extract ephemeral attacker IPs and ingest raw log data into the cognitive pipeline.
- Engineered a deterministic Active Defense Governance Layer in Python to intercept and audit AI-generated actuation commands, enforcing strict infrastructure IP whitelisting to prevent AI hallucinations and host-level Self-Denial-of-Service (DoS).
- Ported the agentic engine to defend a simulated multi-tenant Web Application stack, reconfiguring RAG playbooks and decoders to handle Layer 7 telemetry and triage web-specific vulnerabilities (SQLi/RCE).
- Achieved autonomous threat containment by programmatically actuating dynamic iptables drop rules against validated malicious endpoints, preserving core DNS and network routing integrity.

Secure LLMops Gateway (Stateless Security Proxy)

May 2026

SecOps & Platform Engineer

- Engineered a stateless, high-performance security proxy in Python (FastAPI) designed to intercept, sanitize, and audit Generative AI prompt payloads prior to external routing.
- Designed a high-precision Data Loss Prevention (DLP) engine using regular expressions to auto-redact PII (Social Security Numbers and Credit Card details) on-the-fly, enforcing strict compliance with PCI-DSS and SOC 2 data-in-motion standards.
- Overrode LLM safety-refusal guardrails on redacted payloads by developing a custom context preservation layer in system instructions, allowing local Llama 3.2 models to process sanitized text without false-positive blocks.
- Implemented SIEM-ready JSON logging to stdout that tracks processing latency, transaction UUIDs, and detection flags while completely preventing raw-text PII leakage to downstream log collectors.
- Hardened application delivery by packaging the proxy inside a lightweight Docker container configured to run under a restricted non-root system user (appuser UID 10001) to eliminate container breakout risks.

EXPERIENCE

Karuwaa Express

Sandy, UT

IT Operations Manager

August 2024 – April 2025

- Executed a Formal Risk Treatment Plan for high-risk assets and implemented Role-Based Access Control (RBAC).
- Designed mandatory Staff Security Awareness Training, achieving a 0% phishing click rate.
- Mitigated internal fraud through least privilege enforcement and comprehensive security monitoring.

Intermountain Health

UT

Patient Access (Identity & Compliance)

May 2024 – August 2024

- Enforced HIPAA and PCI-DSS compliance protocols during patient intake processes.
- Mitigated identity theft risks through rigorous PII verification and insurance validation.

Walmart eCommerce

Remote

Partner Support Specialist

September 2020 – September 2021

- Managed enterprise workflows across Salesforce instances, resolving high-priority technical and partner issues.

- Drove process optimization resulting in a 20% improvement in overall team productivity.

CenturyLink

IT Support Help Desk

Monroe, LA

January 2019 – May 2019

- Managed Active Directory for 100+ endpoints, enforcing GPOs and resolving Access Control List (ACL) violations.
- Provided remote systems support and integration troubleshooting for VPN and Citrix infrastructure using PowerShell.

CERTIFICATIONS

CompTIA CySA+ | CompTIA SecAI+ (MSCM Capstone Candidate)